SUPPLEMENTARY INFORMATION

App methodology

1. Spheroid Formation, Treatment, and Image Collection

- Perform 3D tumour spheroid formation, treatments, and image collection using established methods.
- Save the collected images in JPEG or TIFF formats for compatibility with SphereMetrics.

2. Software Installation

- Install R software and the required packages:
  - Download and install R ○ Download and install RStudio ○ Install the required R packages running the following commands:

```
install.packages("shiny")
install.packages("EBImage")
install.packages("openxlsx")
```

- Download the SphereMetrics application:
  - Save the provided R script (SphereMetrics.R) in a known directory on your computer.
  - Open RStudio and navigate to the script's location.
- Launch the Shiny application:
  - Set the working directory to the folder containing the script, then run the app using:

```
shiny::runApp("path/to/your/SphereMetrics.R")
```

○ On Windows, ensure the file path uses either double backslash (\\), example:

```
shiny::runApp("C:\\Users\\YourName\\Downloads\\SphereMetrics.R")
```

○ On Mac, use a regular forward slash path, example:

```
shiny::runApp("/Users/YourName/Downloads/SphereMetrics.R")
```

    - A Shiny web-based interface will launch automatically in your default web browser.

3. Preparing Images for Analysis

- Determine Image Resolution:
    - Find the pixel-to-micron ratio (μm/pixel) from the metadata of the microscope or by consulting the imaging software and system vendor.
- Convert Image Formats:
    - If your images are in a proprietary format, convert them to JPEG or TIFF.

4. Image Analysis Using SphereMetrics

- Launch the SphereMetrics app as described in Step 2.3.
- Upload Images:
    - Click the "Browse" button and select the image files for analysis. Multiple images can be uploaded simultaneously.
- Enter the Pixel-to-Micron Ratio:
    - Input the pixel-to-micron ratio (μm/pixel) in the corresponding field. Ensure all images use the same resolution.
- Choose Threshold Mode:
    - Select Batch Mode to apply the same threshold to all images or Individual Mode to adjust the threshold for each image.
- Adjust Threshold:
    - Use the slider to select a threshold for segmenting spheroids. Adjust the value incrementally using the "+" and "−" buttons, if necessary.
    - To ensure the threshold is selected correctly, the spheroid should be clearly outlined and separated from background objects. If smaller objects remain selected, this will not affect the analysis.
- Analyse Images:

    - Click "Analyse Images" to compute the spheroid area for all uploaded images.
    - The app processes images, applying morphological cleaning to improve segmentation accuracy.
- Navigate Images:
    - Use "Next Image" and "Previous Image" buttons to review the results and verify segmentation for each image.
    - Detected spheroids are highlighted in blue on the preview images.

5. Quality Control

- Preview and Adjust:
    - Check the segmentation quality for each image using the preview display.
    - Adjust thresholds individually for problematic images in "Individual Mode." if selection does not match the spheroid.

6. Exporting Results

- Save Results Table:
    - After analysis completion, the app displays a results table showing spheroid areas (in both pixels and μm²) and eccentricity values. The eccentricity ranges from 0 (a perfect sphere) to 1 (a parabola).
    - The results table will include names of the files in alphabetical order, threshold values, area in pixels and area in μm² and eccentricity.
- Download Results:
    - Click "Download Results" to export the table as an Excel file. The filename will include the current date (e.g., Spheroid_Analysis_2025-01-27.xlsx).

Frequently Asked Questions (FAQs)

---

1. **Why does the app not launch when I run shiny::runApp() in RStudio?**

Ensure that the working directory is correctly set to the location of the SphereMetrics.R file. On macOS or Windows, incorrect file paths or missing dependencies may prevent the app from launching. Confirm that all required packages are installed (including shiny, openxlsx, and EBImage) and that the script has no syntax errors.

---

2. **I receive the error "C stack close to the limit" on macOS—what does this mean and how can I fix it?**

This error indicates a memory stack overflow caused by image size or recursion depth. It is common on macOS due to a lower default stack size. To resolve this, open the Terminal and run the following command before launching RStudio:

```
ulimit -s 16384
open -a RStudio
```

This command increases the stack size limit, allowing the app to run more complex operations without crashing.

---

3. **I am using the same computer and images as a colleague, but I experience slow performance or crashes. Why might this happen?**

System performance can vary due to differences in background processes, available RAM, or package versions. Ensure that no other memory-intensive applications are running. Also verify that you are using the latest versions of R, RStudio, and all necessary packages. Rebooting the machine prior to large batch analysis may also improve performance.

---

4. **The app opens, but no images are visible in the preview window. What could be the issue?**

Check that the uploaded images are in either JPEG or TIFF format, as other formats are not supported. Additionally, ensure the files are not corrupted or zero-byte files. If the images are valid and the problem persists, check for console errors in RStudio that may indicate missing image metadata or unsupported colour modes.

5. **The application processes only part of the dataset or stops responding. How can I avoid this?**

SphereMetrics is optimised for datasets of up to approximately 150 images. Exceeding this number may cause memory issues, particularly on computers with limited RAM. If large-scale analysis is required, consider splitting the dataset into smaller batches and analysing them sequentially.

---

6. **My image contains only one spheroid, but the app fails to identify it accurately. What should I do?**

Certain image conditions—such as low contrast, uneven illumination, or artefacts—may prevent accurate segmentation. In such cases, switch to **Individual Threshold Mode** and manually adjust the threshold using the slider and preview tools until the spheroid is clearly delineated. Avoid over-segmentation that may include background noise.

---

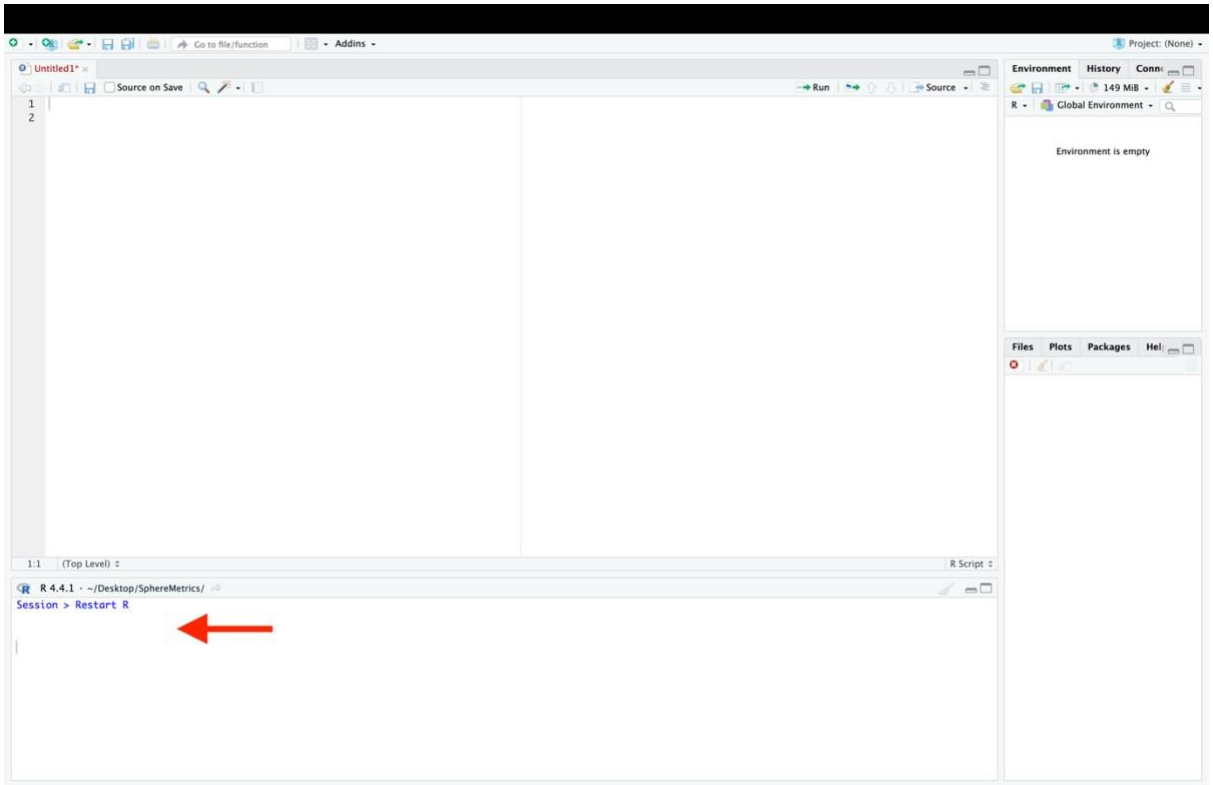
7. **What should I do if the threshold slider has no visible effect on the segmentation output?**

This typically occurs when the image intensity range is very narrow (e.g., very bright or very dark images). Ensure the images are correctly exposed and not oversaturated. If using images from external software, check that they are not pre-processed or compressed in a way that flattens intensity levels.

---

8. **What should I do if app doesn't open after installing required packages?**

Restart R session: In RStudio, use Session > Restart R to ensure a clean environment. Then re-run the script.



If error persists:

Run the script directly: Instead of calling shiny::runApp(), try running the script line by line in the RStudio console to identify any errors during initialisation.

---

9. The application window opens but appears blank or does not load fully. How can I resolve this?

This issue may occur due to local browser settings or firewall restrictions. Try opening the app in a different web browser (e.g., Firefox, Chrome, or Safari). If the problem persists, restart RStudio and re-run the app. Ensure that your system's firewall is not blocking local ports used by Shiny apps.

---

10. Can I run the application on a shared or institutional computer?

Yes, provided R, RStudio, and the necessary packages are installed with sufficient user permissions. However, note that institutional systems may impose memory or security restrictions. It is advisable to consult your IT support team if you encounter persistent installation or runtime issues on managed systems.

---

11. How can I ensure reproducibility when comparing analyses across different users or systems?

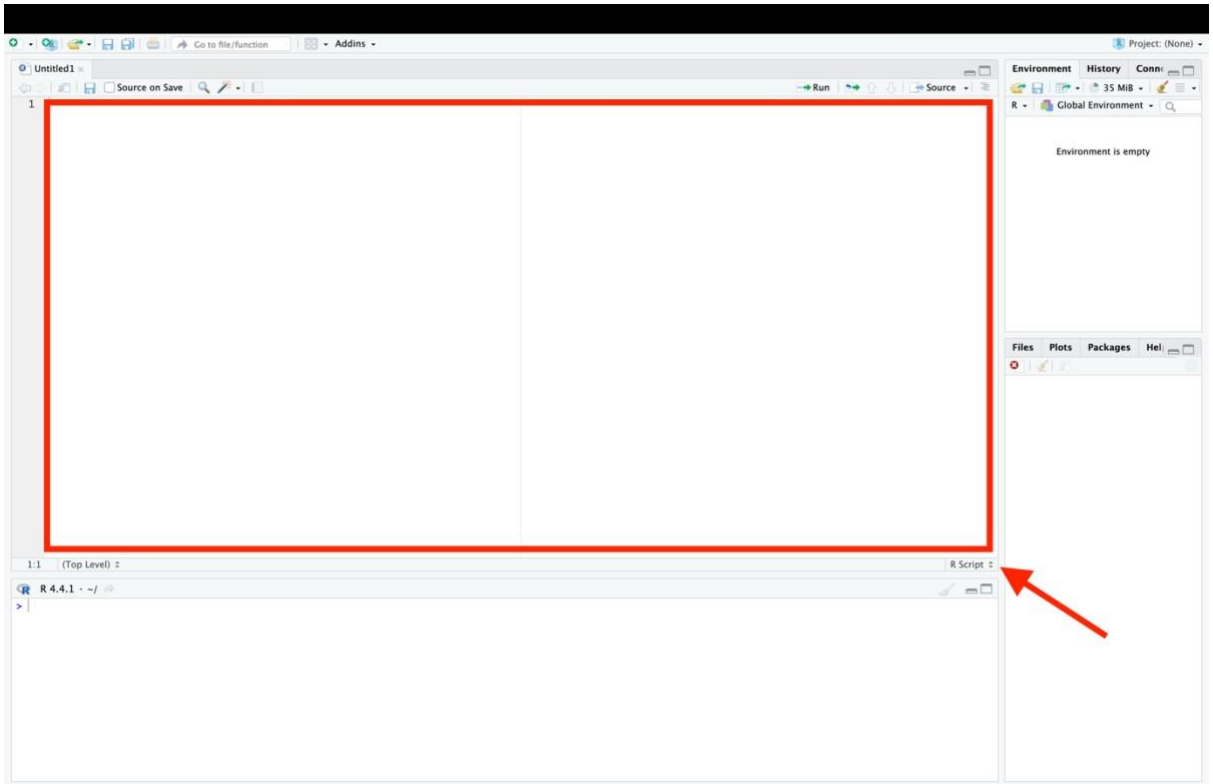
Always ensure that:

- The same image resolution and pixel-to-micron ratio are used.
- Thresholding mode and values are recorded.
- The same version of the SphereMetrics script and R packages are installed.

Exported Excel result files contain threshold values, eccentricity, and area, which can be used for cross-validation across datasets and systems.

12. What version of software should I use?

- **R:** 4.4.1 (2024-06-14) -- "Race for Your Life" or later • **R studio:** Version 2024.04.2+764 (2024.04.2+764) or later

13. Where do I input the code in the R studio?

Input the code as demonstrated below: